**DATA STRUCTURES - IT23302 HW ASSIGNMENT SET - 2**

**Name :** V Rahul

**Roll Number :** 2025115120

**Batch :** IT Batch II Second Year

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**1.Define an integer array. Declare an integer pointer and initialize it to point to the first element of the array. Use pointer arithmetic to traverse the array and print the elements.**

**CODE :**

```
#include<stdio.h>
int main()
{
  int arr[100],n,ind,val;
  int *arrptr;
  printf("Enter the number of elements : ");
  scanf("%d",&n);
  printf("Enter Array Elements \n");
  for(int i=0;i<n;i++) scanf("%d",&arr[i]);
  arrptr=arr;
  printf("Before Modifying\n");
  printf("Traverse the array\n");
  for(int i=0;i<n;i++)
  {
    printf("%d ",*(arrptr+i));
  }
  printf("\nEnter the index to be modified : ");
  scanf("%d",&ind);
  printf("Enter the new element : ");
  scanf("%d",&val);
  *(arrptr+ind)=val;
  printf("After Modifying \n");
 printf("Traverse the array\n");
  for(int i=0;i<n;i++)
  {
    printf("%d ",*(arrptr+i));
  }
  return 0;

}
```

**OUTPUT**

PS C:\My Files\01_ACADEMICS\SEMESTER_3\DSA\hw_2> gcc 1.c -o 1

PS C:\My Files\01_ACADEMICS\SEMESTER_3\DSA\hw_2> gcc 1.c -o 1.exe

PS C:\My Files\01_ACADEMICS\SEMESTER_3\DSA\hw_2> .\1.exe

Enter the number of elements : 3

Enter Array Elements

10

20

40

Before Modifying

Traverse the array

10 20 40

Enter the index to be modified : 2

Enter the new element : 30

After Modifying

Traverse the array

10 20 30

**2. Calculate string length using a pointer.**

**CODE**

```
#include<stdio.h>

int main()
{
  char str[100];
  char* ptr;
  int count=0;
  printf("Enter the String : ");
  fgets(str,sizeof(str),stdin);
  ptr=str;
  for(int i=0;*(ptr+i)!='\0';i++)
  {
    count++;
  }
  printf("String length = %d",count);

}
```

**OUTPUT**

PS C:\My Files\01_ACADEMICS\SEMESTER_3\DSA\hw_2> gcc 2.c -o 2

PS C:\My Files\01_ACADEMICS\SEMESTER_3\DSA\hw_2> gcc 2.c -o 2.exe

PS C:\My Files\01_ACADEMICS\SEMESTER_3\DSA\hw_2> ./2.exe

Enter the String : Rahul

String length = 6

**3.Write a function that calculates both the sum and product of two numbers. Return these two results using pointers passed as arguments to the function.**

**CODE :**

```c
#include<stdio.h>
void func(int ,int ,int*,int*);
int main()
{
  int n1,n2,sum,prod;
  printf("Enter the two numbers : ");
  scanf("%d %d",&n1,&n2);
  func(n1,n2,&sum,&prod);
  printf("Sum = %d\nProduct = %d\n",sum,prod);
  return 0;


}
void func(int n1,int n2,int* sum,int* prod)
{
  *sum=n1+n2;
  *prod=n1*n2;

}
```

**OUTPUT:**

PS C:\My Files\01_ACADEMICS\SEMESTER_3\DSA\hw_2> gcc 3.c -o 3

PS C:\My Files\01_ACADEMICS\SEMESTER_3\DSA\hw_2> gcc 3.c -o 3.exe

PS C:\My Files\01_ACADEMICS\SEMESTER_3\DSA\hw_2> ./3.exe

Enter the two numbers : 3 4

Sum = 7

Product = 12

**4.Create an array of structures. Use dynamic memory allocation to allocate memory space and use a structure pointer with pointer arithmetic to iterate and access/modify members.**

**CODE**

```
#include<stdio.h>
#include<stdlib.h>
struct student
{
  int roll;
  char name[100];
  int marks;
};
int main()
{
  struct student *stud;
  struct student *ptr;
  int n;
  printf("Enter the number of students : ");
  scanf("%d",&n);
  stud=(struct student*)calloc(n,sizeof(struct student));
  for(int i=0;i<n;i++)
  {
  printf("Enter Student %d Details(Roll,name,marks)\n",i+1);
  scanf("%d",&(*(stud+i)).roll);
  scanf("%s",(*(stud+i)).name);
  scanf("%d",&(*(stud+i)).marks);
  }
  ptr=stud;
  printf("Accessing Student details \n");
  for(int i=0;i<n;i++)
  {
  printf("Student %d Details(Roll,name,marks)\n",i+1);
  printf("%d\n",(*(ptr+i)).roll);
  printf("%s\n",(*(ptr+i)).name);
  printf("%d\n",(*(ptr+i)).marks);
  }
  return 0;


}
```

**OUTPUT**

PS C:\My Files\01_ACADEMICS\SEMESTER_3\DSA\hw_2> gcc 4.c -o 4

PS C:\My Files\01_ACADEMICS\SEMESTER_3\DSA\hw_2> ./4

Enter the number of students : 2

Enter Student 1 Details(Roll,name,marks)

2025115120

V.RAHUL

100

Enter Student 2 Details(Roll,name,marks)

2025103029

RJ

100

Accessing Student details

Student 1 Details(Roll,name,marks)

2025115120

V.RAHUL

100

Student 2 Details(Roll,name,marks)

2025103029

RJ

100

**5.Define two functions with the same signature. Declare a function pointer and use it to call both functions.**

**CODE**

```
#include<stdio.h>
int sum(int n1,int n2)
{
  return n1+n2;
}
int prod(int n1,int n2)
{
  return n1*n2;
}
int main()
{
  int n1,n2;
  int(*fp)(int,int);

  printf("Enter the two numbers : ");
  scanf("%d %d",&n1,&n2);
  fp=sum;
  printf("Sum = %d\n",((fp)(n1,n2)));
  fp=prod;
  printf("Product = %d",((fp)(n1,n2)));
  return 0;


}
```

**OUTPUT**

PS C:\My Files\01_ACADEMICS\SEMESTER_3\DSA\hw_2> gcc 5.c -o 5

PS C:\My Files\01_ACADEMICS\SEMESTER_3\DSA\hw_2> ./5

Enter the two numbers : 10 20

Sum = 30

Product = 200

**6.Declare an integer, an integer pointer, and a double pointer. Print the values and addresses at each level.**

**CODE**

```
#include<stdio.h>
int main()
{
  int n,*ptr,**dptr;
  printf("Enter the number : ");
  scanf("%d",&n);
  ptr=&n;
  dptr=&ptr;
  printf("Value level\n");
  printf("Value of n= %d\n",n);
  printf("Address of n= %d\n",&n);
  printf("Single pointer level\n");
  printf("Value of ptr = %d\n",ptr);
  printf("Address of ptr = %d\n",&ptr);
  printf("Value of n using ptr = %d\n",*ptr);
  printf("Double pointer level\n");
  printf("Value of dptr = %d\n",dptr);
  printf("Address of dptr = %d\n",&dptr);
  printf("Value of ptr using dptr = %d\n",*dptr);
  printf("Value of n using dptr = %d\n",**dptr);
  return 0;
}
```

**OUTPUT**

PS C:\My Files\01_ACADEMICS\SEMESTER_3\DSA\hw_2> ./6

Enter the number : 10

Value level

Value of n= 10

Address of n= -1533019540

Single pointer level

Value of ptr = -1533019540

Address of ptr = -1533019552

Value of n using ptr = 10

Double pointer level

Value of dptr = -1533019552

Address of dptr = -1533019560

Value of ptr using dptr = -1533019540

Value of n using dptr = 10

**7.Sort an array of floating-point numbers using Selection Sort.**

**CODE**

```
#include<stdio.h>
int main()
{
  float arr[100],temp;
  int n,i,j,min;
  printf("Enter the number of elements : ");
  scanf("%d",&n);
  for(i=0;i<n;i++)
    scanf("%f",&arr[i]);
  printf("Floating type Array before sorting \n");
  for(i=0;i<n;i++)
    printf("%f ",arr[i]);
  printf("Selection sort\n");
  for(int i=0;i<n;i++)
  {
    min=i;
    for(int j=i+1;j<n;j++)
    {
      if(arr[j]<arr[min]) min=j;
    }
    temp=arr[i];
    arr[i]=arr[min];
    arr[min]=temp;

  }
  printf("Floating type Array after sorting \n");
  for(i=0;i<n;i++)
    printf("%f ",arr[i]);




}
```

**OUTPUT**

PS C:\My Files\01_ACADEMICS\SEMESTER_3\DSA\hw_2> ./7

Enter the number of elements : 5

80

50

90

40

30

Floating type Array before sorting

80.000000 50.000000 90.000000 40.000000 30.000000 Selection sort

Floating type Array after sorting

30.000000 40.000000 50.000000 80.000000 90.000000

**8.Define a structure Student with id, name, and grade. Create an array of student structures and sort them based on different field**

**CODE**

```
#include<stdio.h>
#include<stdlib.h>
#include<string.h>
struct Student{
  char name[100];
  int reg;
  float grade;
};
void sortByName(struct Student *stud, int n);
void sortByReg(struct Student *stud, int n);
void sortByGrade(struct Student *stud, int n);
void display(struct Student *stud, int n);

int main()
{
  struct Student *stud;
  int n;
    int choice;
  printf("Enter the number of students : ");
  scanf("%d",&n);
  stud=(struct Student*)calloc(n,sizeof(struct Student));
  printf("Enter student details : ");
  for(int i=0;i<n;i++)
  {
    printf("Enter Student %d (name,roll,grade)",i+1);
    scanf("%s",stud[i].name);
    scanf("%d",&stud[i].reg);
    scanf("%f",&stud[i].grade);
  }

    do{
      printf("Before Sorting\n");
      for(int i=0;i<n;i++)
  {

    printf("%s ",stud[i].name);
    printf("%d ",stud[i].reg);
    printf("%f\n",stud[i].grade);
  }
      printf("Sort by:\n");
```

```
    printf("1. Name\n");
    printf("2. ID\n");
    printf("3. Grade\n");
    printf("Enter choice(-1 to exit): ");
    scanf("%d",&choice);
    switch(choice)
    {
      case 1:
      sortByName(stud,n);
      display(stud,n);
      break;
      case 2:
      sortByReg(stud,n);
      display(stud,n);
      break;
      case 3:
      sortByGrade(stud,n);
      display(stud,n);
      break;
      default:
      printf("Invalid choice");
    }
    }while(choice!=-1);
    return 0;
}
void sortByName(struct Student* stud,int n)
{
  struct Student temp;
  for(int i=0;i<n;i++)
  {
    for(int j=0;j<n-i-1;j++)
    {
      if(strcmp(stud[j].name,stud[j+1].name)>0)
      {
        temp=stud[j];
        stud[j]=stud[j+1];
        stud[j+1]=temp;

      }
    }
  }
}
void sortByReg(struct Student* stud,int n)
{
  struct Student temp;
```

```
  for(int i=0;i<n;i++)
  {
    for(int j=0;j<n-i-1;j++)
    {
      if(stud[j].reg>stud[j+1].reg)
      {
        temp=stud[j];
        stud[j]=stud[j+1];
        stud[j+1]=temp;

      }
    }
  }
}
void sortByGrade(struct Student* stud,int n)
{
  struct Student temp;
  for(int i=0;i<n;i++)
  {
    for(int j=0;j<n-i-1;j++)
    {
      if(stud[j].grade>stud[j+1].grade)
      {
        temp=stud[j];
        stud[j]=stud[j+1];
        stud[j+1]=temp;

      }
    }
  }
}
void display(struct Student *stud,int n)
{
  printf("After sorting \n");
    for(int i=0;i<n;i++)
  {

    printf("%s ",stud[i].name);
    printf("%d ",stud[i].reg);
    printf("%f\n",stud[i].grade);
  }
  printf("\n--------------END--------------\n");

}
```

**OUTPUT**

PS C:\My Files\01_ACADEMICS\SEMESTER_3\DSA\hw_2> ./8

Enter the number of students : 3

Enter student details : Enter Student 1 (name,roll,grade)A

1000

9

Enter Student 2 (name,roll,grade)B

2000

8

Enter Student 3 (name,roll,grade)C

500

10

Before Sorting

A 1000 9.000000

B 2000 8.000000

C 500 10.000000

Sort by:

1. Name

2. ID

3. Grade

Enter choice(-1 to exit): 1

After sorting

A 1000 9.000000

B 2000 8.000000

C 500 10.000000

--------------END--------------

Before Sorting

A 1000 9.000000

B 2000 8.000000

C 500 10.000000

Sort by:

1. Name

2. ID

3. Grade

Enter choice(-1 to exit): 2

After sorting

C 500 10.000000

A 1000 9.000000

B 2000 8.000000

--------------END--------------

Before Sorting

C 500 10.000000

A 1000 9.000000

B 2000 8.000000

Sort by:

1. Name

2. ID

3. Grade

Enter choice(-1 to exit): 3

After sorting

B 2000 8.000000

A 1000 9.000000

C 500 10.000000

--------------END--------------

Before Sorting

B 2000 8.000000

A 1000 9.000000

C 500 10.000000

Sort by:

1. Name

2. ID

3. Grade

Enter choice(-1 to exit): -1

Invalid choice

**9.Sort a set of negative numbers using radix sort**

**CODE**

```
#include<stdio.h>
int main()
{
  int bucket[10][100],bucketcount[10];
  int arr[100],n,i,j,k,rem,nop=0,divisor=1,max,min,pass,offset;
  printf("Enter the number of elements : ");
  scanf("%d",&n);
  printf("Enter Array Elements \n");
  for(i=0;i<n;i++)
  {
    scanf("%d",&arr[i]);

  }
max=arr[0];
min=arr[0];
for(i=0;i<n;i++)
{
  if(arr[i]<min) min=arr[i];
  if(arr[i]>max) max=arr[i];

}
if(min<0) offset=-min;
else offset=0;
for(i=0;i<n;i++)
{
arr[i]+=offset;
}
max+=offset;
while(max>0)
{
  nop++;
  max/=10;
}
for(pass=0;pass<nop;pass++)
{
  for(i=0;i<10;i++) bucketcount[i]=0;
  for(i=0;i<n;i++)
{
rem=arr[i]/divisor%10;
bucket[rem][bucketcount[rem]]=arr[i];
bucketcount[rem]++;
```

```
}
i=0;
for(k=0;k<10;k++)
{
  for(j=0;j<bucketcount[k];j++)
  {
    arr[i]=bucket[k][j];
    i++;
  }
}
divisor *= 10;

}
for(i=0;i<n;i++)
{
arr[i]-=offset;
}


  printf("Array after sorting \n");
  for(i=0;i<n;i++)
    printf("%d ",arr[i]);

  return 0;
}
```

**OUTPUT**

PS C:\My Files\01_ACADEMICS\SEMESTER_3\DSA\hw_2> ./9

Enter the number of elements : 4

Enter Array Elements

9

-2

8

-7

Array after sorting

-7 -2 8 9

**10.Write a C program to copy one array to another using pointers**

**CODE**

```
#include<stdio.h>
int main()
{
  int arr[100],copy[100],n;
  int *arrptr;
  printf("Enter the number of elements : ");
  scanf("%d",&n);
  printf("Enter Array Elements \n");
  for(int i=0;i<n;i++) scanf("%d",&arr[i]);
  arrptr=arr;
  printf("Displaying from the copied Array\n");
  for(int i=0;i<n;i++)
  {
    copy[i]=*(arrptr+i);
    printf("%d ",copy[i]);
  }
  return 0;
}
```

**OUTPUT**

PS C:\My Files\01_ACADEMICS\SEMESTER_3\DSA\hw_2> ./10

Enter the number of elements : 4

Enter Array Elements

1

2

5

7

Displaying from the copied Array

1 2 5 7